

Project Diary — Seenu

Enrollment A2345923074, Roll 74. Group 298, B.Tech 7CSE (Evening), Amity School of Engineering and Technology. Project MAESTRO — Design and Evaluation of a Safe Multi-Agent System for Natural Language-Driven Desktop Task Automation. Guide: Dr. Rajni Sehgal Kaushik. My role in the team is research and evaluation. I am responsible for the literature review, for the comparison of the systems that already exist, for the research gap, and for deciding how our own system will be measured once it is finished.

Week 1

The first week was mainly about finding out what field we were standing in. Before any paper could be read sensibly we had to know which areas the project touches, and I worked through that until we had five: artificial intelligence, natural language processing, multi-agent systems, desktop automation and AI safety. Naming them properly mattered, because each one has its own vocabulary and its own set of venues, and searching without knowing that wastes a lot of time. I ran the first searches on Google Scholar and arXiv to see how much work exists in each area and how recent it is, and it was clear immediately that the desktop agent work is very new and moving quickly. I also did the economic side of the feasibility study, listing every tool we intend to use along with its licence and its free limits, which is how we can honestly write zero against the bill of material.

Week 2

This was the reading week. I collected around thirty papers and, rather than reading them in the order I found them, I set up a matrix first with columns for the year, the approach to planning, whether the system executes on a real machine, what benchmark it uses, and what it reports. Filling a table forces you to notice what a paper leaves out, which turned out to be the most useful decision I made this month. On the agent side I read the work on reasoning combined with acting, on task planning with tools, on open-ended agents that learn skills for themselves, and on frameworks where several specialised agents cooperate instead of one large prompt doing everything. From all of that I wrote the first part of the Literature Review chapter, organised by theme rather than one paragraph per paper, because a list of summaries is not a review.

Week 3

This week the study moved to the systems people actually use, and I ran the comparative study while Jairaj tested the same three systems on the machine. I compared them property by property rather than feature by feature, and the properties I chose are the ones that matter for our argument: how an action is represented before it runs, whether there is any classification of danger, whether the user gets a preview, whether anything can be undone, whether there is a real audit trail, and whether the system defends itself against instructions hidden inside a file or

a web page. The pattern was the same in every column. Capability is strong and improving. Everything to do with safety is either missing or is a dialog box. The finding I keep coming back to is the last row of my table, which is that none of these systems report a safety measurement at all, so there is no way to say that one is safer than another.

Week 4

I wrote the research gap chapter this week, and I was careful to make every claim in it something a reader can check against my own matrix rather than something they have to accept on trust. The gap came out in four parts. Existing work measures how capable an agent is and not how safe it is. A confirmation popup is an interface feature with no testable properties, whereas a typed plan with fixed scoring rules can actually be examined. Resistance to instructions hidden in documents is never reported as a headline number for desktop systems. And there is no public dataset of instructions paired with plans for this task, least of all one carrying safety labels. Writing the gap also pushed me into thinking about measurement early, because if we are going to claim our system is safer then the word has to mean a number, and I began sketching what those numbers should be.

Week 5

The requirement work this week needed real situations rather than a wish list, so I wrote the user scenario analysis. I built three personas, the student who is not a programmer and whose files are scattered everywhere, the developer who wants speed but also wants a log he can check afterwards, and the evaluator who is looking for method and honest numbers rather than a polished demo. Then I wrote out the situations each of them would actually put to the system, which is what produced the six categories of task the requirements now cover. In parallel I started defining the measurements properly, and the important one is the unsafe execution rate, which counts how often a consequential action ran without the confirmation it should have had. Our target for it is zero. I also decided it must always be reported together with the rate at which the system asks for confirmation unnecessarily, because a system that asks about everything would score zero on the first number while being useless.

Where I stand

The literature review, the comparative study and the research gap are done and written, and five weekly reports are filed. What comes next is mine as well: the evaluation methodology chapter, where the hundred benchmark tasks are designed and each measurement is defined formally, followed by the seed dataset with its annotation guidelines and an agreement score calculated on a shared portion so that the labelling can be shown to be consistent.